



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO NAVEEN AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

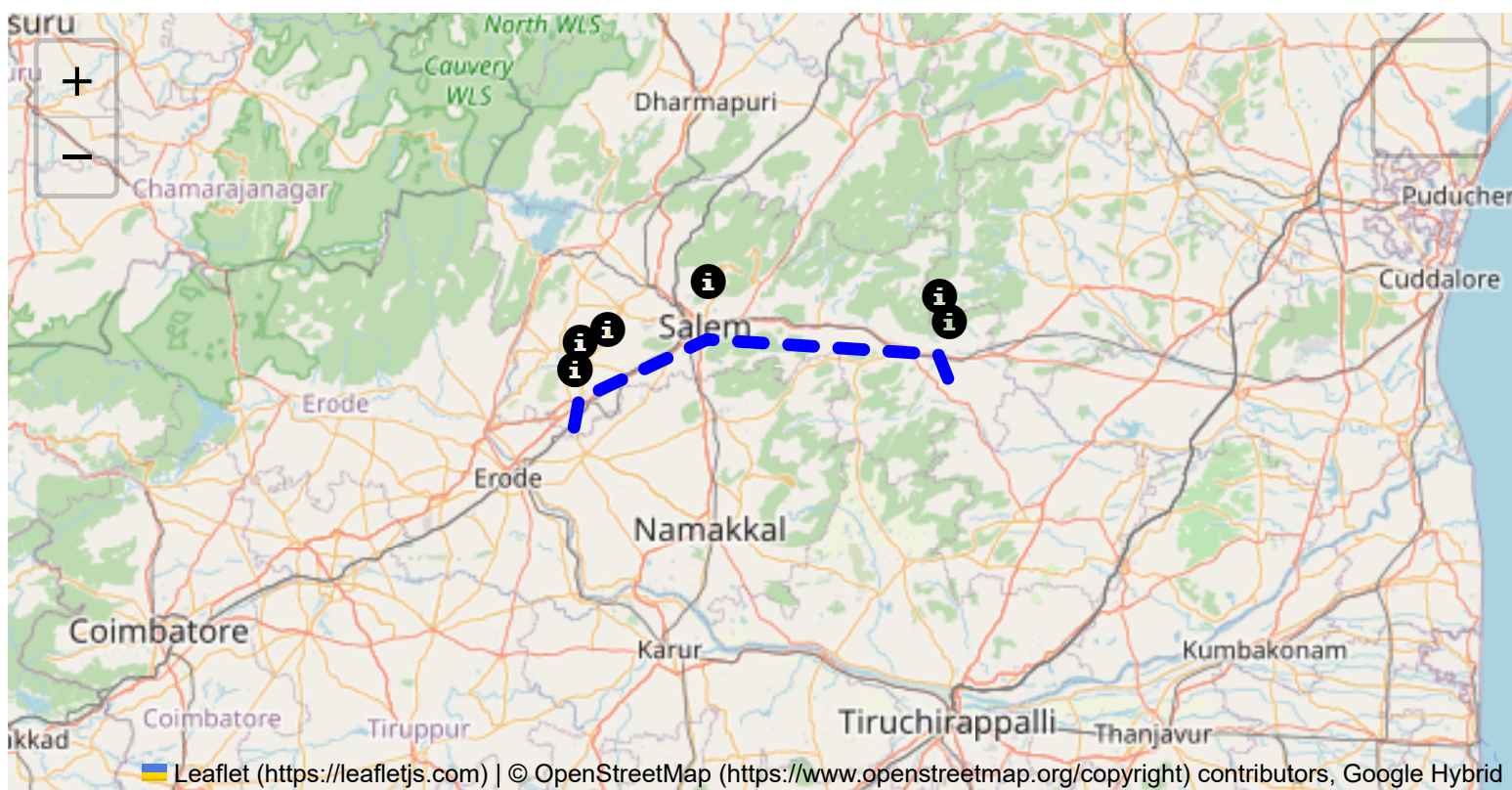
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

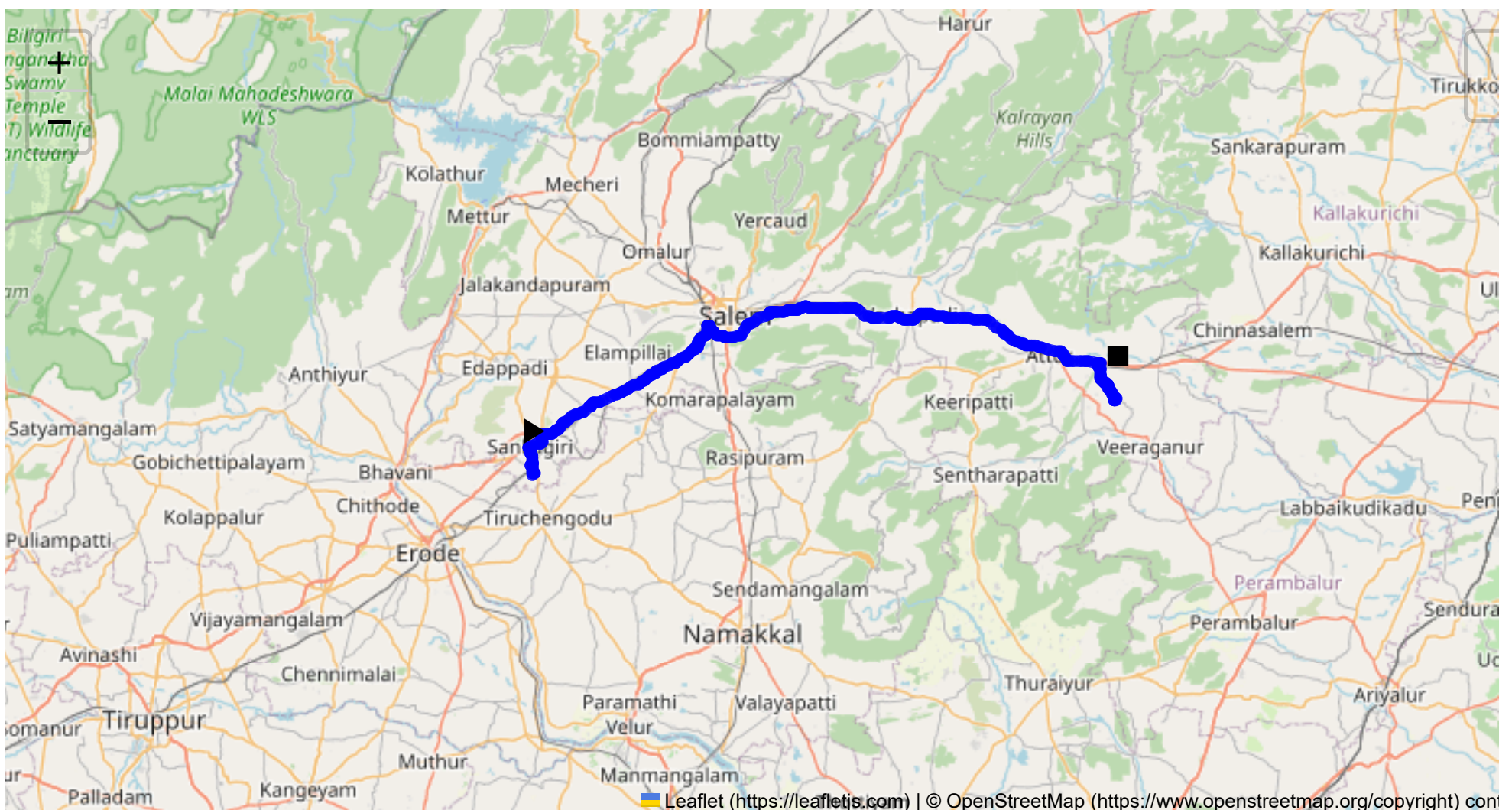
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 112.71 km
Estimated Duration: 2.2 hours
Adjusted Duration (Heavy Vehicle): 2.7 hours
Start: (11.4381, 77.8734)
End: (11.5394, 78.68898)



Welcome to the Journey Risk Management Study

1. **Overview of the Route Map:** The route from CVQF+23W, Sangagiri to GMQQ+X6H, Salem spans approximately 112.71 kilometers, following a path through Puthur, Salem - Kochi Hwy in Sangagiri, Vaikundam, K R Nagar, and Anna Nagar, Vadachennimalai. This route primarily utilizes both state highways and local roads, passing through urban and semi-urban areas.

2. **Typical Weather Conditions and Potential Weather-Related Hazards:** The region generally experiences a tropical climate with hot summers, heavy south-west and north-east monsoons, and moderate winters. The monsoon season (June to September and October to November) can lead to heavy rainfall, which may cause flooding, landslides in hilly areas, and reduced visibility, impacting travel time and safety.
3. **Analysis of Traffic Patterns:** Traffic is typically more congested during peak hours (8-10 AM and 5-8 PM) in urban sections near Salem and Sangagiri. The NH 544 junctions and Salem - Kochi Highway are known for heavy traffic, particularly with commercial vehicles, which could cause delays.
4. **Assessment of Road Quality and Infrastructure:** Road quality varies; major highways are generally well-maintained but local roads may have potholes and narrower lanes. Infrastructure improvements in urban areas may cause temporary detours or slowdowns. Care should be taken at intersections, especially those unregulated by traffic signals.
5. **Suggestions for Alternative Routes for Emergencies:** For diversions, NH 44 can be used as an alternative to avoid congestion-prone areas. Local bypass roads in Salem can also provide alternate paths, though they might extend travel time.
6. **Summary of Local Regulations Affecting Hazardous Material Transport:** Tamil Nadu follows national guidelines for the transport of hazardous materials, requiring permits, appropriate vehicle labeling, and adherence to speed limits. Transport should preferably avoid peak hours and sensitive areas such as schools and hospitals.
7. **Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:** While specific data must be verified locally, historically, roads in this region report incidents due to speeding and driver fatigue, particularly involving heavy trucks on highways. Compliance with regulations and proper training are crucial for safety.
8. **Environmental Considerations and Sensitive Areas:** The route passes near several environmentally sensitive zones and urban communities. Care should be taken to avoid spillages, especially near water bodies, agricultural land, and forest patches to minimize environmental impact.
9. **Analysis of Communication Coverage:** Most major telecom services provide good coverage throughout the route; however, rural sections might have intermittent connectivity. Drivers should carry updated maps offline and be prepared for potential communication gaps.
10. **Estimated Emergency Response Times:** Urban areas like Salem can expect quicker emergency response times (15-30 minutes), while rural stretches may see delays of up to an hour depending on proximity to emergency services and road conditions.
11. **Overall Summary of Risk Assessment:** This route presents moderate risks related to weather, congestion, and road conditions. Proper scheduling to avoid peak hours, meticulous adherence to traffic and safety regulations, and preparedness for weather-related challenges are recommended. Emergencies should prompt swift diversions, and all communications equipment should be tested for reliability before embarking on the route. Regular checks on vehicle conditions and adherence to regulatory guidelines for hazardous material transport will further ensure safety and compliance.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43961, 77.87341	15 KM/Hr
2	Turn	High	11.44024, 77.87528	15 KM/Hr
3	Turn	High	11.44898, 77.87410	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	Medium	11.46392, 77.87328	30 KM/Hr
6	Turn	Medium	11.47391, 77.86844	30 KM/Hr
7	Turn	High	11.47412, 77.86848	15 KM/Hr
8	Turn	Medium	11.48361, 77.88779	30 KM/Hr
9	Turn	Medium	11.49167, 77.88772	30 KM/Hr
10	Turn	High	11.49440, 77.88480	15 KM/Hr
11	Turn	Medium	11.49412, 77.88450	30 KM/Hr
12	Blind Spot	Blind Spot	11.49383, 77.88442	10 KM/Hr
13	Turn	Medium	11.49412, 77.88450	30 KM/Hr
14	Turn	High	11.49440, 77.88480	15 KM/Hr
15	Turn	Medium	11.49365, 77.88596	30 KM/Hr
16	Turn	Medium	11.53588, 77.95823	30 KM/Hr
17	Turn	Medium	11.64151, 78.11805	30 KM/Hr
18	Turn	Medium	11.64199, 78.11844	30 KM/Hr
19	Turn	Medium	11.64222, 78.11898	30 KM/Hr
20	Turn	Medium	11.64205, 78.12104	30 KM/Hr
21	Turn	High	11.64139, 78.12168	15 KM/Hr
22	Turn	Medium	11.63984, 78.12086	30 KM/Hr
23	Turn	Medium	11.63869, 78.12103	30 KM/Hr
24	Turn	Medium	11.59320, 78.62069	30 KM/Hr
25	Blind Spot	Blind Spot	11.59137, 78.66635	10 KM/Hr
26	Turn	High	11.59099, 78.66630	15 KM/Hr
27	Turn	High	11.59101, 78.66563	15 KM/Hr
28	Turn	Medium	11.57258, 78.66759	30 KM/Hr
29	Turn	High	11.57036, 78.66783	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
30	Turn	High	11.57027, 78.66863	15 KM/Hr
31	Turn	Medium	11.57000, 78.66881	30 KM/Hr
32	Turn	Medium	11.56791, 78.66867	30 KM/Hr
33	Turn	Medium	11.56688, 78.66914	30 KM/Hr
34	Turn	Medium	11.56113, 78.67725	30 KM/Hr
35	Turn	High	11.56034, 78.67714	15 KM/Hr
36	Turn	High	11.56015, 78.67751	15 KM/Hr
37	Turn	Medium	11.55524, 78.67911	30 KM/Hr
38	Turn	High	11.54254, 78.68881	15 KM/Hr
39	Turn	High	11.53979, 78.68804	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
7	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
8	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
10	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
14	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
19	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
20	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
22	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
23	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
24	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
25	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
26	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
27	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
28	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
29	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
30	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
34	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
36	clinic	Lavanya Clinic	11.652121, 78.4795146	30 km/h	Medium
37	hospital	Amsh Agathiyar Sidha Hospital	11.650564, 78.483542	30 km/h	Medium
38	clinic	Sudha Polyclinic	11.647825, 78.508537	30 km/h	Medium
39	hospital	Government Hospital, Kothampadi	11.62641, 78.542199	30 km/h	Medium
42	hospital	Government Hospital	11.5617651, 78.6762531	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
13	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
15	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
21	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
31	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
32	school	Thangam ZEE public school	11.6653353, 78.251641	30 km/h	Medium
33	college	AVS College od Arts and Science, Salem.	11.6642672, 78.2584134	30 km/h	Medium
35	college	Mahendra Engineering College	11.6672537, 78.2641303	30 km/h	Medium
40	school	Vani Vikas Matric Hr.sec school	11.6212713, 78.5462658	30 km/h	Medium
41	school	Saraswati Metric Hr sec School	11.5910827, 78.6402264	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45007, 77.87201



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46392, 77.87328



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47391, 77.86844



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47412, 77.86848



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48361, 77.88779



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49167, 77.88772



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49440, 77.88480



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49412, 77.88450



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49383, 77.88442



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49412, 77.88450



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49440, 77.88480



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49365, 77.88596



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.53588, 77.95823



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64151, 78.11805



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64199, 78.11844



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64222, 78.11898



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64205, 78.12104



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64139, 78.12168



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63984, 78.12086



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63869, 78.12103



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59320, 78.62069



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.59137, 78.66635



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59099, 78.66630



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59101, 78.66563



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57258, 78.66759



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57036, 78.66783



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57027, 78.66863



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57000, 78.66881



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56791, 78.66867



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56688, 78.66914



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56113, 78.67725

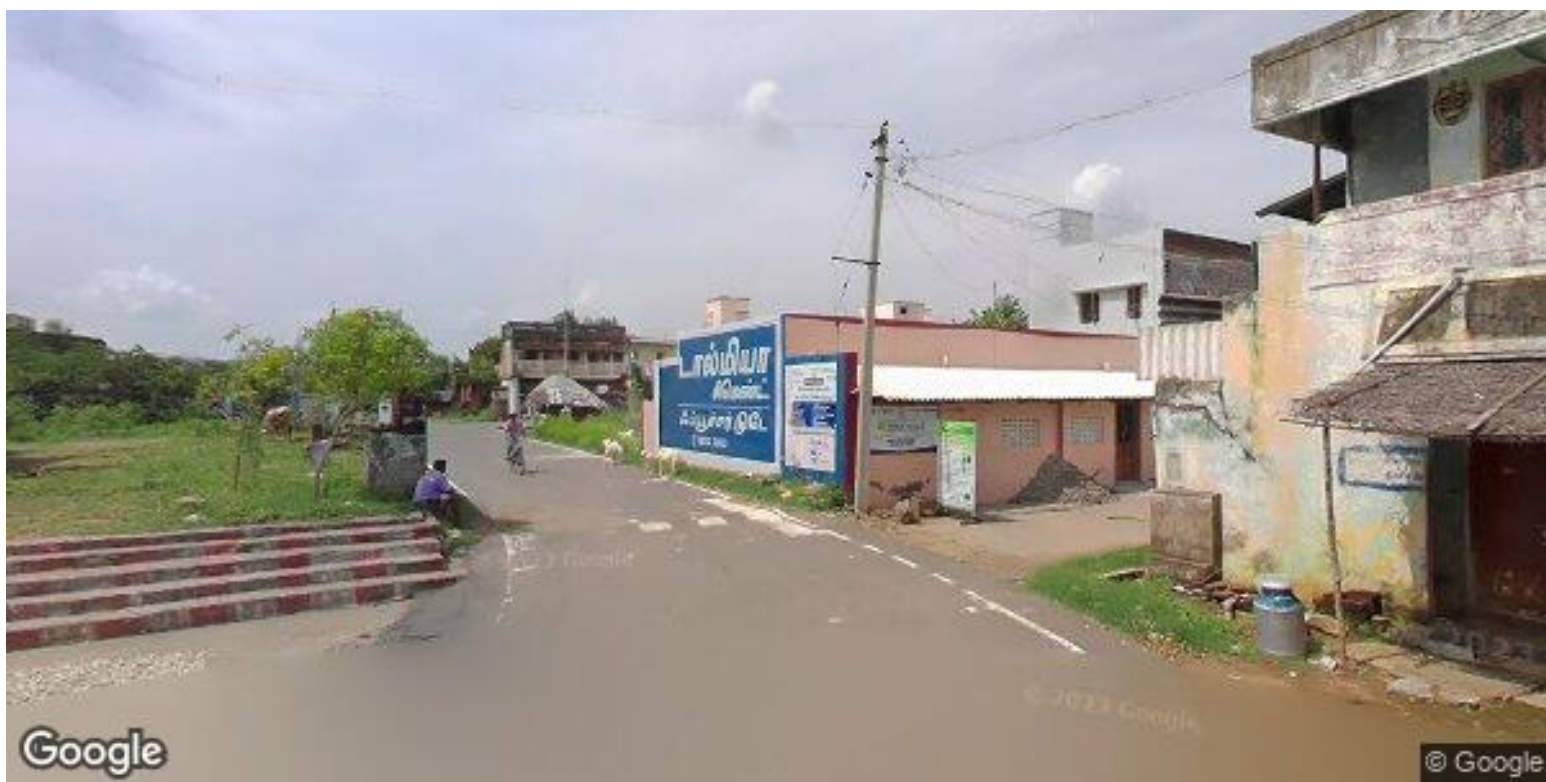


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.56034, 78.67714



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.56015, 78.67751



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55524, 78.67911



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.54254, 78.68881



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.53979, 78.68804
